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Research on Lightweight PCB Defect Detection
Method

Final Thesis

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Abstract

To address the high computational cost of existing PCB defect detection models, this paper proposes a lightweight improved YOLOv8 model combined with GhostNet. By replacing the original backbone network, the model parameters and computation are effectively reduced while maintaining detection accuracy. Experiments on a public PCB defect dataset show that the proposed model greatly reduces model size and parameters with only minor accuracy loss. It also achieves obvious speed improvement on the CPU, especially for large and simple defects. The overall model realises a good balance between lightweight performance and detection effect, making it more suitable for real-time industrial inspection and edge deployment.

Acknowledgements

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Thanks to the school and the provided learning platform and experimental environment, I have had the opportunity to explore cutting-edge technologies such as deep learning and machine vision, and apply theoretical knowledge to practical engineering problems. At the same time, I would like to express my gratitude to all the teachers who have imparted knowledge in the course. Their rigorous academic attitude and professional academic literacy have laid a solid foundation for my graduation project.

Finally, I would like to express my deepest gratitude to my family. They have always provided me with unconditional understanding, support, and encouragement, which is the most solid backing for me to complete my studies with peace of mind and focus on scientific research.

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List of Acronyms

CNN	Convolutional Neural Network
AOI	Automatic optical inspection
CPU	Central Processing Unit
DFL	Distribution Focal Loss
FPGA	Field-Programmable Gate Array
GPU	Graphics Processing Unit
mAP	Mean Average Precision
mAP@0.5	mean Average Precision at IoU threshold 0.5
PCB	Printed Circuit Board
PAFPN	Path Aggregation Feature Pyramid Network
Swin	Shifted Window (Swin Transformer)
ViT	Vision Transformer
YOLO	You Only Look Once
YOLOv8	You Only Look Once version 8
GhostNet	Lightweight CNN architecture based on Ghost convolution

Chapter 1

Introduction

1.1 Motivation, Aims and Objective

Printed circuit boards (PCBs) have always been indispensable components of various electronic devices. With the continuous expansion of scale and improvements in precision in the electronic manufacturing industry, the demand for PCBs is also constantly increasing. In the actual PCB industrial production process, there are six types of process defects with high occurrence rates, which are missing holes, mouse bites, open circuits, short circuits, spurs, and spurious copper. These types of defects are small in size and difficult to detect, causing circuit breaks, signal disturbances, and, in severe cases, potential safety hazards to equipment operation [1]. Therefore, the level of defect detection directly determines the overall reliability of electronic products. A common type of detection method in current production is manual visual screening and automatic optical detection. The problem with these two methods is that they rely on manual assistance for judgment, which inevitably leads to low detection efficiency, insufficient flexibility in applicable scenarios, and lower accuracy for small hidden defects [1]. In order to solve the problems caused by manual detection, deep learning-based object detection technology has been continuously developing in recent years. The existing mainstream models have successfully improved the accuracy and speed of detection, but the models generally have a large number of parameters, and there are still problems, such as insufficient speed and expensive computational consumption in industrial applications [4]. Another type of lightweight model proposed focuses on smaller model volume and faster model speed, but this type of model often comes with a decrease in detection accuracy. In the actual production line of PCB manufacturing enterprises, online automatic optical inspection (AOI) equipment only needs to determine whether the PCB is a good product. Defective circuit boards are analyzed by professional engineers to determine the cause of the defect. Therefore, from

the perspective of engineering deployment, the main task of the front-end inspection model on the production line is not to classify all six types of defects in detail, but to use lightweight models to achieve extremely fast and reliable screening.

Based on the above practical problems, the core research direction of this article is to build a PCB defect detection model that balances lightweight and detection accuracy. While ensuring the accuracy of defect recognition remains unchanged, the model size is reduced, and the image inference speed is improved. In addition, this study will compare the impact of different defects on the detection accuracy and computational speed of lightweight models, in order to better assess the performance of models on edge devices and meet the needs of industrial sites. The specific research content is as follows. Firstly, obtain a publicly available dataset containing six typical PCB defects, and optimise and transform the YOLOv8 backbone network using GhostNet's lightweight structure to reduce model parameters and computational complexity. After the model training is completed, a comparative experiment is conducted to evaluate the advantages and disadvantages of the lightweight model compared to the base model in terms of detection accuracy, model size, and inference speed. Finally, this study will test the recognition performance of the model for different types of defects and verify the practicality of the method in practical industrial scenarios.

1.2 Literature Review

In the current practical application of PCB defect detection, single-stage detection algorithms are widely used due to their fast inference speed, which better meets the real-time quality inspection needs of industrial production lines. In 2021, Zhang et al. attempted to use MobileNetV2 to construct a lightweight backbone network while incorporating a dual-channel attention mechanism and PAFPN feature fusion structure. This modification successfully simplified the model structure and improved the recognition performance of the model for minor defects extensively [4]. In the following year, Wu et al. embedded the Ghost lightweight module into the YOLOv5 network. They used Ghost convolution to reduce redundant feature maps and combined it with an attention mechanism to ensure that the detection accuracy did not significantly decrease, ultimately achieving optimisation of model size and inference speed [5].

Researchers have also made considerable explorations in accurately identifying small defects. Starting from 2021, the first improved model is based on YOLOv4. The model adopts Mosaic data augmentation and Mish activation function, which effectively improves the anti-interference ability of the model in complex backgrounds [3]. However, the improved model still has a larger volume and cannot effectively solve the problem of lightweighting. In 2022, the coordinate attention mechanism was introduced into the YOLOX model, which emphasised the model's ability to extract subtle defect features. The improved model structure is lightweight and can be used for embedded device deployment [6]. In 2023, Yu et al. designed a new multi-scale feature pyramid structure and adaptive localisation loss function to further enhance the model's ability to capture small defects, especially improving the detection accuracy of low feature recognition defects [7].

In addition to traditional convolutional neural network models, Transformer-based detection methods have also been gradually applied to PCB defect detection in recent years. An and Zhang built the LPViT Visual Transformer model in 2022, which enhances the stability of PCB defect classification and detection through global feature association

modelling [8]. In the same year, Chen et al. abandoned the traditional CNN backbone network and replaced it with the Swin Transformer structure, relying on a hierarchical attention mechanism to synchronously extract global semantic information and local detail features. They achieved good detection accuracy on publicly available test datasets [9]. In 2023, Yang and Kang attempted to integrate the SwinV2 Transformer module into the YOLOv7 network. Their research successfully enhanced the model's feature extraction ability for complex morphological defects, but significantly increased the number of model parameters and computational burden, making it difficult to deploy in practice [10]. It can be confirmed that there is still room for improvement and promising prospects in the research of basic transform detection methods.

After reviewing and analyzing existing relevant research, I have found that there are still several urgent issues that need to be addressed in the aspect of PCB defect detection. Firstly, models with high detection accuracy often have complex structures and high resource consumption, making it difficult to adapt to edge terminal devices with limited computing power and meet the deployment needs of industrial sites. Secondly, the lightweight transformation of most models can lead to a decrease in detection accuracy [7]. This problem is more prominent in the detection of minor defects, which limits the practical application of lightweight models. Finally, most existing research focuses more on precision testing in laboratory environments, and there are few lightweight practical testing solutions that can be used for industrial production scenario design, which makes it difficult for defect detection technology to truly be implemented in industrial quality inspection.

Based on the research gaps mentioned above, this article will construct a new PCB defect detection model using the GhostNet lightweight structure and YOLOv8 network as the foundation. The core idea is to compress the model lightweight, reduce the number of parameters and computation, while preserving the network's feature learning ability as much as possible, balancing the relationship between detection accuracy, inference speed, and device deployability. I hope this study can make up for the shortcomings of existing

research and find a more reasonable direction for the application on industrial production lines.

Chapter 2

Methodology and Results

2.1 Methodology

This article proposes a printed circuit board defect detection model that uses GhostNet lightweight backbone instead of redundant feature maps in the traditional YOLOv8 object detection framework. The overall research methodology and process consist of the following parts. After obtaining the public dataset and grouping it, first complete the construction and debugging of the Ultralytics basic model, and then replace and debug the Ghost model. In the experiment, identical datasets and variable settings were used for comparative testing, and the comprehensive performance of the two models was evaluated and analysed. This article uses YOLOv8n as the benchmark model, which achieves an ideal balance between detection speed and positioning accuracy compared to other YOLO versions. It is suitable as a starting point for lightweight transformation in edge deployment scenarios. The lightweight transformation mainly involves replacing the standard convolutional layers in the YOLOv8n backbone feature extraction network with Ghost convolutional modules one by one. The modified Ghost model retains the original network and detection head structure of YOLOv8, without introducing additional attention mechanisms or deepening the network hierarchy. The goal is to build a lightweight PCB defect detection system that can run on edge devices with limited computing power.

The experiment used a publicly available PCB defect dataset, and the images were taken from real industrial quality inspection sites. The six typical defects included in the dataset are shown in Figure 2.1.1.

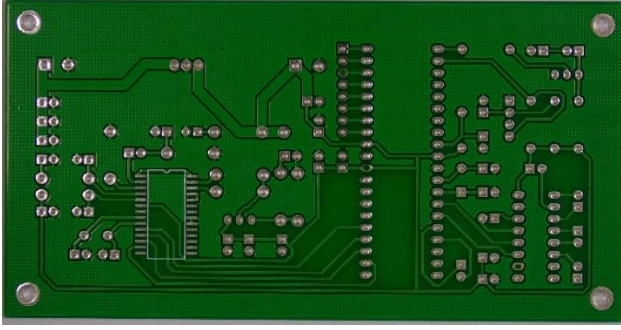


Figure 2.1.1 (a) Missing hole

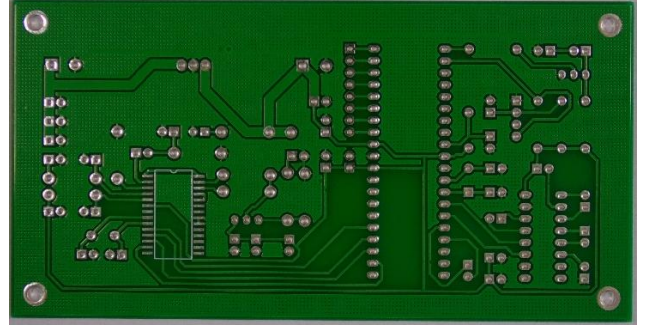


Figure 2.1.1 (b) Mouse bite

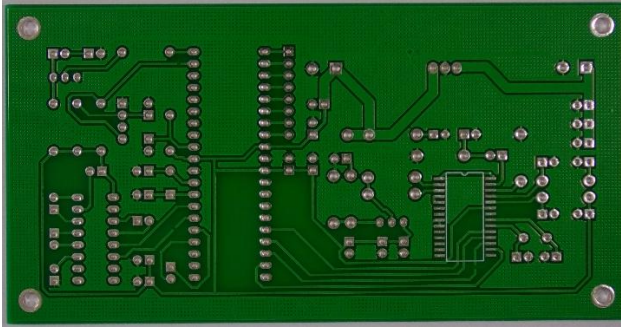


Figure 2.1.1 (c) open circuit

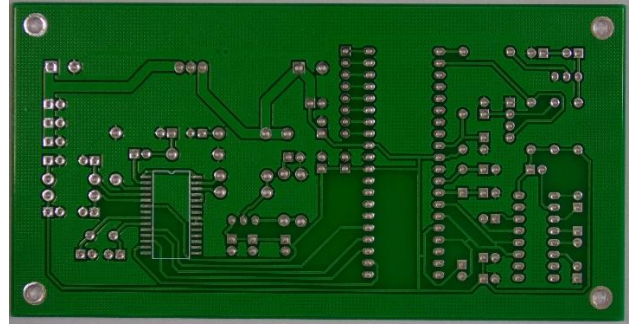


Figure 2.1.1 (d) short circuit

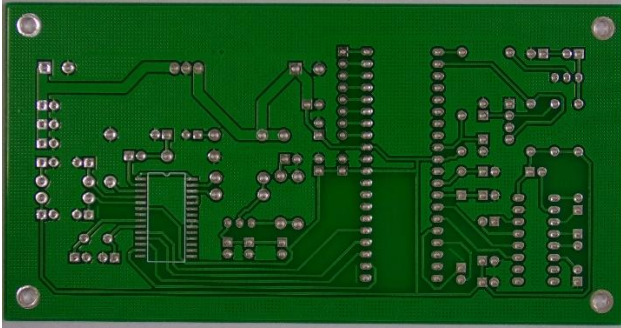


Figure 2.1.1 (e) spur

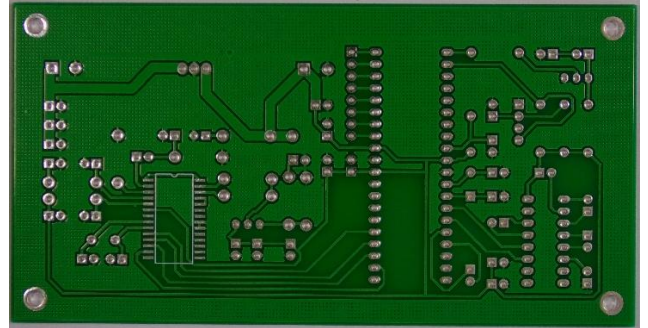


Figure 2.1.1 (f) spurious_copper

Figure 2.1.1: Six basic defects in the dataset

Figure 2.1.1 (a) shows the missing hole, which refers to the failure to penetrate the conductive holes or mounting holes that should have existed on the circuit board, resulting in the interruption of electrical connections. Figure 2.1.1 (b) shows the mouse bite, which refers to an irregular notch as defects appear at the edge of the guideline, usually caused by abnormalities in the etching process. Figure 2.1.1 (c) open circuit and Figure 2.1.1 (d) short circuit refer to the phenomenon of short circuit and open circuit in the circuit. Figure

2.1.1 (e) spur that is sharp and excessive copper foil protruding from the edge of a wire, which affects the accuracy and safety of the circuit. If the process cleaning is not thorough, it will result in irregular excess copper spots remaining on the board, that is Figure 2.1.1 (f), spurious copper [2]. The above-mentioned defects have significant harm but are relatively small and difficult to detect, and the selected dataset contains a large number of small defects with a target area of less than 1% of the total pixels in the image. These defects can not carry much pixel information, so they require the detection model to have high positioning accuracy and feature extraction capability. Therefore, this dataset can effectively test the true performance of the model on small-sized object detection tasks. The number of images with six different types of defects in this dataset is basically equal, and the image size and pixels are also the same, which is beneficial for control experiments and obtaining more accurate and comprehensive data.

Before model training, it is necessary to divide the dataset into three parts, following the common practice in the field of deep learning object detection [4]. The percentage of the training set, validation set, and test set is 7:1.5:1.5. At the early stage of training, I had not realised the importance of the test set, so the result is very good, and the Ghost model performs nearly as accurately as the basic model. After my supervisor reminded me to add the test set, there was a significant difference in accuracy between the two models, and the results became more accurate, realistic, and valuable for discussion. Because the number of images with different defects is roughly the same, stratified sampling was not used when dividing the dataset; rather, random sampling was directly applied. Ultimately, it can also ensure that the distribution ratio of the six types of defects in the three subsets is roughly balanced, which can reflect the performance differences of different categories and will not interfere with the stability of training and the reliability of evaluation results.

This article selects YOLOv8n as the initial model for defect detection. This version has undergone specialised parameter compression and structural simplification for lightweight application scenarios, with the least number of parameters and the fastest inference speed in the entire series. It can balance detection speed and positioning accuracy, so it is suitable

as a basic framework for further lightweight transformation towards industrial edge deployment. The backbone network of the model mainly undertakes the task of feature extraction, extracting multi-scale visual features layer by layer from the input image. The deepening of network hierarchy leads to a gradual decrease in spatial resolution of feature maps, forming a hierarchical feature expression from low-level texture edges to high-level semantic concepts. Meanwhile, YOLOv8n adopts a feature pyramid structure to receive feature maps from different levels of the backbone network. Through a top-down upsampling path and horizontal connection feature fusion, it effectively combines shallow high-resolution positional information with deep strong semantic category information. Finally, YOLOv8 adopts an anchor-free detection method to directly predict the centre point coordinates and width to height offsets of the target box, eliminating the manual setting of hyperparameters such as anchor size and aspect ratio in traditional anchor box methods. The anchor-free box detection method simplifies the training process and reduces the complexity of parameter tuning. This feature makes YOLOv8n particularly suitable for handling the PCB micro defect detection task of interest in this article. Although YOLOv8n is already relatively lightweight, it is still difficult to deploy on edge devices with limited computing and storage resources. Therefore, this article introduces Ghost convolution to implement a lightweight transformation of YOLOv8n's backbone feature extraction network.

Researchers have found that in deep convolutional neural networks, there are a large number of feature maps with similar content between the layer-by-layer generated feature maps. They call it ghost convolution. These ghost convolutions also use expensive convolution operations. If these less important ghost feature maps can be replaced, the model's parameter count and computational cost will be further reduced. Based on the above situation, Ghost convolution divides the generation process of output feature maps into two stages. In the first stage, conventional convolutions with fewer channels are used to generate intrinsic feature maps, which carry the main feature information of the layer. In the second stage, a series of lower computational cost linear transformations are applied to the intrinsic feature maps to generate corresponding shadow feature maps. Finally, the

intrinsic feature map and shadow feature map are concatenated in the channel dimension to form a complete result consistent with the traditional convolution output dimension. This feature generation method replaces the calculations that originally required standard convolution with inexpensive linear operations. Under the condition of keeping the output dimension unchanged, the total number of parameters and floating-point operations in the model is greatly reduced. The intrinsic feature maps themselves are still extracted through convolution operations, and only removing redundant feature maps maintains a high level of non-linear feature expression ability for the entire network [5].

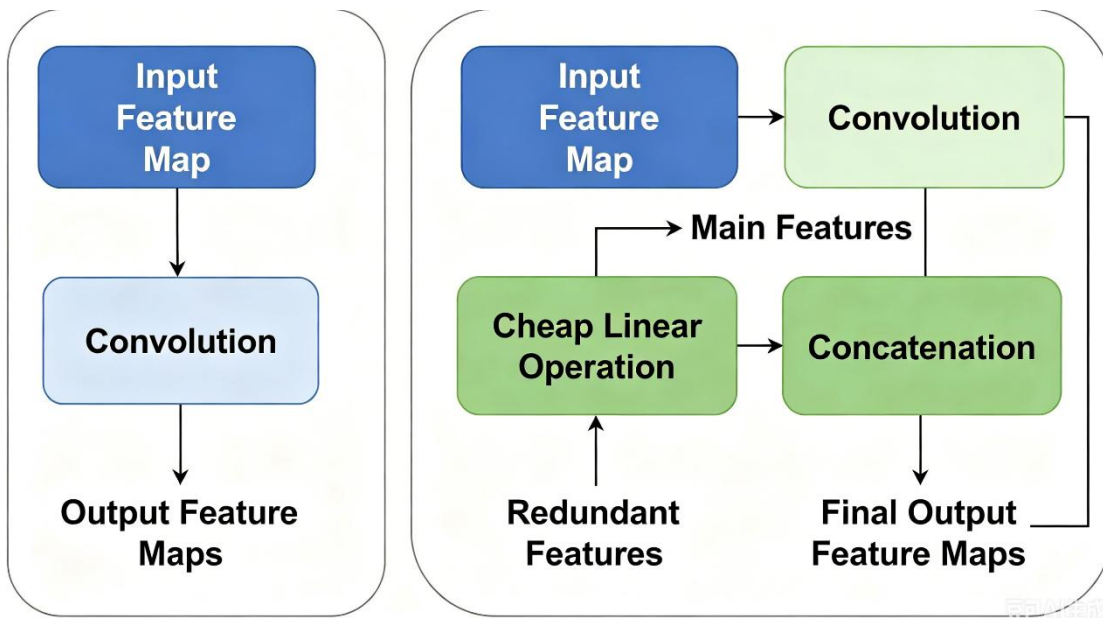


Figure 2.2.2 Comparison of Basic Model and Ghost Model

The Ghostnet model used in this article is based on the Ghost convolution idea mentioned above, and is a lightweight modified on the YOLOV8N backbone network. This model fully preserves the original neck network and detection head structure of YOLOv8, without any additional lightweight operations. At the same time, no additional attention mechanism is introduced to avoid introducing new parameters and computational modules to offset the efficiency benefits brought by Ghost convolution, which avoids failing to obtain intuitive comparison results.

All models in this article were trained and evaluated in the same software and hardware environment. This article chooses to be based on CPU rather than GPU. In addition to the configuration of my computer itself, it is also because although CPU training takes longer and the effect is slightly inferior to GPU, it does not have graphics processor acceleration, which is actually more similar to the actual deployment conditions with limited computing power on the simulation edge. In terms of software, this article chooses PyCharm as the development software, PyTorch as the computation engine, and Ultralytics as the object detection training tool. The input image resolution for all models is $640 * 320$. AdamW optimiser is used to achieve more stable convergence performance and better generalisation performance during the training process. During model training, YOLOV8's built-in early stop is enabled as the training termination condition. The model on the validation set $mAP@0.5$. If there is no effective improvement in the indicator within 50 rounds, the training will be automatically terminated in advance, and the model from the previous 50 rounds will be saved as the optimal model. The early stopping mechanism can effectively prevent overfitting of the model on the training set, while avoiding the continuous consumption of ineffective computing resources. The more rounds a model goes through during training, the better. Excessive training can result in the model performing almost perfectly on the training set but deteriorating on the test set. This is overfitting, in layman's terms, it means that the model memorises the training questions too hard, even treating the errors and irrelevant traces of right and wrong in the questions as test points, and as a result, it won't be able to do it on a new test paper. This results in overfitting models performing poorly when encountering completely unfamiliar defects in actual production scenarios. Therefore, it is necessary to use early stopping and datasets with more diversity and images to prevent overfitting.

After the training of the two models is completed, this article evaluates the detection accuracy and model lightweighting based on commonly used evaluation standards in the field of industrial defect detection. $MAP@0.5$ is the average accuracy mean when the intersection to union ratio threshold is set to 0.5, which can intuitively reflect the

comprehensive detection accuracy. The parameter count directly reflects the complexity of the model and the storage space it consumes, and is the core data for measuring the effectiveness of lightweight transformation. While the size of the model file is closer to the storage resource consumption in actual deployment scenarios. In addition to the inference time output from the test set, the single inference time is also an important indicator for evaluating whether the model meets the real-time detection requirements. In addition to the main evaluation indicators mentioned above, the data automatically output by models such as Map50-95, box_loss also have research and analysis value.

In addition to using the overall mAP index to measure the comprehensive detection ability of the model, this article conducted statistical and comparative analysis on the detection accuracy of the model on six different types of defects. There are significant differences in the difficulty of detecting different types of defects due to their appearance, scale range, and contrast with the background. Larger and more obvious defect types, such as missing holes, often have higher detection accuracy, while smaller and more blurry defect types, such as spur, have relatively lower detection accuracy. By comparing the accuracy performance of defects with different detection difficulties on lightweight models and basic models, we can identify weak links in detection capabilities and optimise them accordingly in subsequent research. For example, increasing the proportion of such defects in the dataset and conducting targeted data augmentation for such defects. This classification comparison method can better reflect the characteristics of lightweight models and prepare for subsequent parameter tuning.

When evaluating the lightweight performance of object detection models, the average single inference speed output by the model is usually used as the evaluation criterion for speed. However, in practical industrial quality inspection scenarios, a noteworthy but relatively less discussed issue in the literature is whether the category of defects contained in the image will affect the inference time of the model. In theory, the computational complexity of the model depends on the size of the input image and the network structure itself, and is not directly related to the types and quantities of defects in the image.

To verify the theoretical inference mentioned above, this paper conducted a statistical experiment on the inference time grouped by category on the test set. The specific approach is to randomly select the same number of pictures from six different types of defects in the test set, input the same model to perform reasoning, record and calculate the average reasoning time of each different defect. If the percentage difference in time consumption between groups is relatively small, it can be concluded that the defect category does not constitute an actual factor affecting inference speed. The practical significance of this analysis is that if the experimental results determine the class independence of inference time, it ensures that there is no need to switch models and parameters due to defect types on the production line during actual deployment. On the contrary, if a certain type of defect is observed to have significantly higher or lower inference time, it suggests that the defect can be preprocessed or provide ideas for further lightweighting.

2.2 Results

Both models were successfully trained and obtained results. The experimental results will be discussed below. After five hundred rounds of training, the basic YOLO model was successfully trained. It is on the validation set of the validation set mAP@0.5. The indicators did not show significant improvement from around 470 rounds of training; no early stop was triggered. The ghost model triggers an early stop mechanism after 394 rounds of training, so YOLO automatically outputs 344 rounds as the best train. In the subsequent discussion of this article, the YOLO basic model trained for five hundred rounds and the Ghost model trained for 344 rounds will be compared and discussed. Both models were able to successfully detect defects and correctly label their locations.

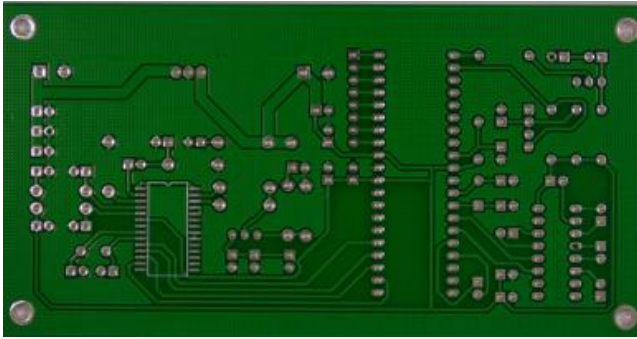


Figure 2.2.1 (a) input of yolov8n

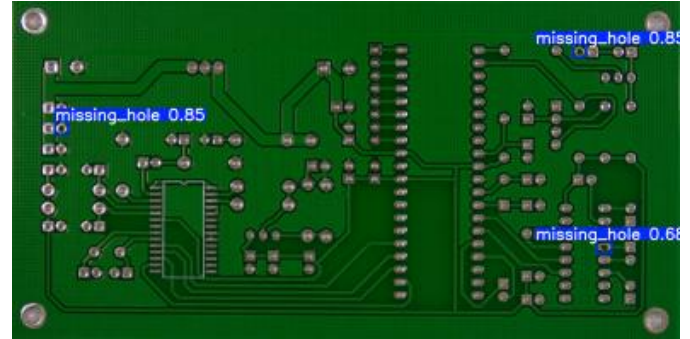


Figure 2.2.1 (b) output of yolov8n

Figure 2.2.1 Detection of a single image using the basic YOLO model

Figure 2.2.1 shows one randomly picked defect figure from the test set and its detection result output by the basic yolo model, while Figure 2.2.2 shows one randomly picked defect figure from the test set and its detection result output by the Ghost model.

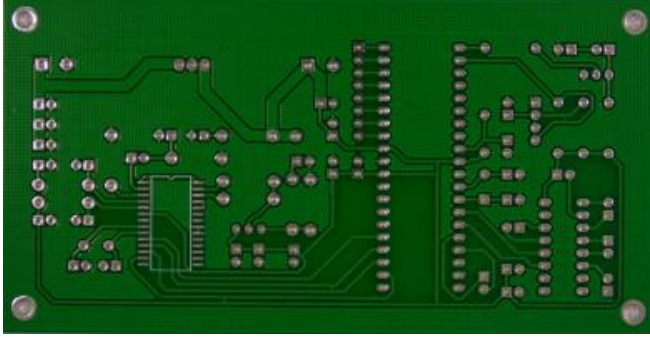


Figure 2.2.2 (a) input of the Ghost model

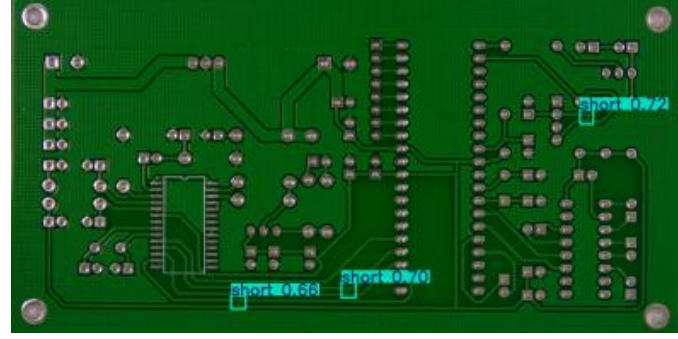


Figure 2.1.2 (b) output of Ghost model

Figure 2.2.2 Detection of a single image using the Ghost model

From the training process, both models showed a trend of steady convergence. In the first few rounds of training, the model weights are still in a random initialisation state, resulting in low detection accuracy. As the number of rounds increases, the accuracy on the validation set steadily climbs, while the training loss and validation loss decrease synchronously, which is a common process in deep learning model training. In the middle and later stages of training, the basic YOLOv8n converges faster and ultimately achieves higher convergence accuracy with more convolutional operation units. The GhostNet model, due to the reduction of feature maps in the network structure, has a certain impact on its feature learning ability, resulting in slower convergence and slightly lower final accuracy. But what is satisfactory is that its entire convergence process is stable, without any oscillations, vanishing gradients, or sudden drops in accuracy. This also indicates that the improved model has stability, which is conducive to reproduction and practical application. Meanwhile, after 17.7 hours of training, the lightweight model showed a significant improvement compared to the basic model's 23 hours. This also has benefits such as shortening the R&D cycle, saving time and computing power in actual project development.

The comprehensive performance statistics of the two models are shown in Table 2.2.1

Metric	YOLOv8n	GhostNet	Change
mAP@0.5	82.3%	80.6%	↓1.7%
Parameters	3.15M	1.75M	↓44.4%
Model Size	6.2MB	3.7MB	↓40.3%
CPU Inference Time (per image)	41.9ms	33.6ms	↑20.1%

Table 2.2.1 comprehensive performance statistics of the two models

From the table, it can be seen that after the lightweight transformation of Ghost convolution, the accuracy of the model has decreased. Its mAP decreased from 82.3% to 80.6%, a decrease of 1.7 percentage points. This decrease is actually not small, and it is higher than my expectation of a decrease of less than 1.5% in the proposal. In the context of industrial quality inspection, a precision loss of less than two percentage points is completely acceptable, as the environment and lighting fluctuations on the production line themselves also have certain errors. Ghost model 80.6% map@0.5 It can also meet the basic accuracy requirements in industrial scenarios. This is sufficient to demonstrate that the Ghost model did not compromise the backbone's feature extraction capability.

Compared to the acceptable loss of detection accuracy, the lightweight optimization of the model is satisfactory. The parameter size of the model has been compressed from 3.15M to 1.75M, with a compression ratio of 44.44%, almost halved. The model weight file has also been reduced from 6.2MB to 3.7MB, resulting in a direct decrease of 40.32% in storage space usage. The simplification in both aspects is close to 50%, which is a significant improvement in lightweighting. This level of simplification has greatly expanded the hardware range that the model can be applied to. This can be said to be the

most intuitive result in this experiment. Memory and storage are no longer the bottleneck of PCB defect detection models for edge device applications.

The real-time detection speed of the model has also been significantly improved. Ghost convolution significantly reduces redundant computation in the network, which shortens the average time for the model to process a 640×320 image from 41.9ms to 33.6ms. The increased speed is nearly twenty per cent. This may enable the model to better match fast-paced automated production lines. By processing several more images per second, over time, the overall throughput of the production line will be significantly improved.

Based on these basic defect detection data, we have come to an overall conclusion. The GhostNet lightweight improvement scheme adopted in this article has indeed achieved a significant reduction in the computational overhead and memory usage of the model while sacrificing detection accuracy within an acceptable range, and effectively improved the computing speed. It did not pursue the precision of the last one or two percentage points in the laboratory, but found a more reasonable balance between algorithm performance and engineering practicality.

In addition to the basic data mentioned above, the precision (BoxP) and recall (BoxR) at the bounding box level are also worth paying attention to. Both can assess the impact of lightweight transformation on the model's localisation ability and indicate missed or false detections. The training logs show that the BoxP and BoxR of the original YOLOv8n are 0.905 and 0.755, respectively, while the corresponding values of GhostNet YOLOv8 are 0.899 and 0.725, both of which have decreased.

BoxP only decreased by 0.006, a decrease of less than one per cent, which can be considered as a level within the fluctuation range. This indicates that the lightweight model has not become more prone to misidentifying normal areas as defects. Considering that every false inspection on the industrial production line will result in ineffective manual re-inspection, the stability of BoxP is a positive signal. BoxR decreased from 0.755 to 0.725,

a decrease of 0.030, a decrease of about four percentage points. The absolute value of the change is not significant, but the decrease is not small; it is already five times the decrease of BoxP. This suggests that the lightweight model may not be significantly affected in finding the right defects, but may be somewhat affected in finding all defects, although 0.725 is still an acceptable value. This may be because Ghost convolution compresses feature channels while causing some weak defect feature information to be lost in shallow networks, causing the model to ignore these targets and directly resulting in missed detections rather than false detections.

This trend is not good. Missing inspections may pose safety hazards, product recall risks, and other losses. Misinspections often result in high re-inspection rates and wasted production capacity. The problems caused by missed detections are often more serious than false detections under normal circumstances, but false detections are equally serious in scenarios where deep learning PCB inspection is applicable, such as fully automated unmanned factories. The leakage of the lightweight model in this study is within an acceptable range and is acceptable in conventional civilian PCB quality inspection scenarios. However, for fields such as aerospace and medical electronics that have zero tolerance for leakage, their applicability needs to be carefully evaluated.

There is no significant difference in the loss function, box loss and CLS loss between the two models. This indicates that there is no significant loss in the ability of the lightweight model to select the position of the box and distinguish the category of defects. The problem in the above discussion lies in whether defects can be found, rather than the statement of accuracy.

The Dfl loss of lightweight models slightly increases, with smaller defects and blurred edges. Therefore, subsequent discussions will focus on the impact of different types of defects with varying sizes on accuracy.

Defect Type	YOLOv8n mAP@0.5 (%)	GhostNet mAP@0.5 (%)	absolute difference (Δ, %)	Relative decrease proportion (%)
Missing hole	99.2	98.5	0.7	0.71
Mouse bite	70.1	67.6	2.5	3.57
Open circuit	83.8	81.5	2.3	2.74
Short	89.5	87.7	1.8	2.01
Spur	77.9	75.4	2.5	3.21
Spurious copper	74.5	72.8	1.7	2.28

Table 2.2.2 The absolute value and percentage of difference in accuracy decrease between YOLO and Ghostnet

From Table 2.2.2, it can be seen that there are significant differences in the impact of lightweight retrofitting on different types of defects.

The detection accuracy of leakage defects has decreased from 99.2% to 98.5%, with an absolute difference of only 0.7 percentage points and a relative decrease of less than 1%. This result is unexpectedly good. Leakage itself is also a small-sized defect, theoretically affected by feature compression. However, the size of the leakage hole is larger than some other small defects, and the leakage hole usually appears as a completely blank circular area on the PCB surface, with extremely high contrast with the surrounding circuits. Although the features are small, they are relatively obvious and easy for the human eye to

distinguish. The prominent defect of this core feature information is not easily lost during feature channel compression in lightweight networks. The high stability of leakage holes also indicates that the performance degradation of lightweight models is not simply proportional to the size of defects, but is closely related to whether the defects are obvious.

The accuracy of open circuit defects and short circuit defects slightly decreased after lightweight modification. The accuracy of open circuit and short circuit decreased by 2.3 and 1.8 percentage points, respectively, with a relative decrease of 2% to 3%. These two types of defects usually have a large target area on the PCB surface and clear defect contours. The lightweight model still retains strong recognition ability. The performance of the two types of defects is basically the same, with a slightly greater decrease in accuracy for open circuits than for short circuits. Open circuits on the board surface may manifest in various forms, such as notches, cracks, etc., which require a higher generalisation ability of the model. The shape of the short circuit is relatively regular, mostly consisting of bridging excess copper foil, with a more singular and stable feature. The overall difficulty of both is not high, and it is reasonable that the detection accuracy of the open circuit is slightly lower than that of the short circuit.

The detection accuracy of mouse bites, burrs, and miscellaneous copper has significantly decreased. The absolute accuracy decrease of these three types of defects ranges from 1.7 to 2.5 percentage points, with a relative decrease of 2.28% to 3.57%. The common characteristics of these three types of defects are that the target size is too small, the edge contour is irregular, and the differentiation from the PCB board background is low. Ghost convolution compresses the number of feature maps, and small defects themselves require more feature maps to describe. Therefore, when the Ghost model reduces redundant feature maps, it is more likely to affect small defects.

By cross-checking the above results with the positioning of the bounding box, it can be observed that. This indicates that the main performance loss of lightweight models is concentrated in the process of finding defects. The slight increase in Box Loss and DFL

Loss further supports this judgment. The positioning ability of the model, especially its fine judgment ability on fuzzy edges, is directly affected by lightweight compression. Once the model successfully locates the defect, it can classify it correctly. Overall, the cost of lightweight transformation is to make the model ignore a very small number of weak feature defects.

From the perspective of industrial applications, among the relatively obvious defects of precision decline mentioned above, mouse bites and burrs are usually tolerable minor defects under conventional civilian PCB quality inspection standards, and the influence of copper impurities is also easy to avoid in actual production. The three types of defects, namely leakage, open circuit, and short circuit, have a direct impact on circuit functionality. These happen to be the categories that maintain the best accuracy in lightweight models, ensuring their practicality in civilian PCB quality inspection scenarios.

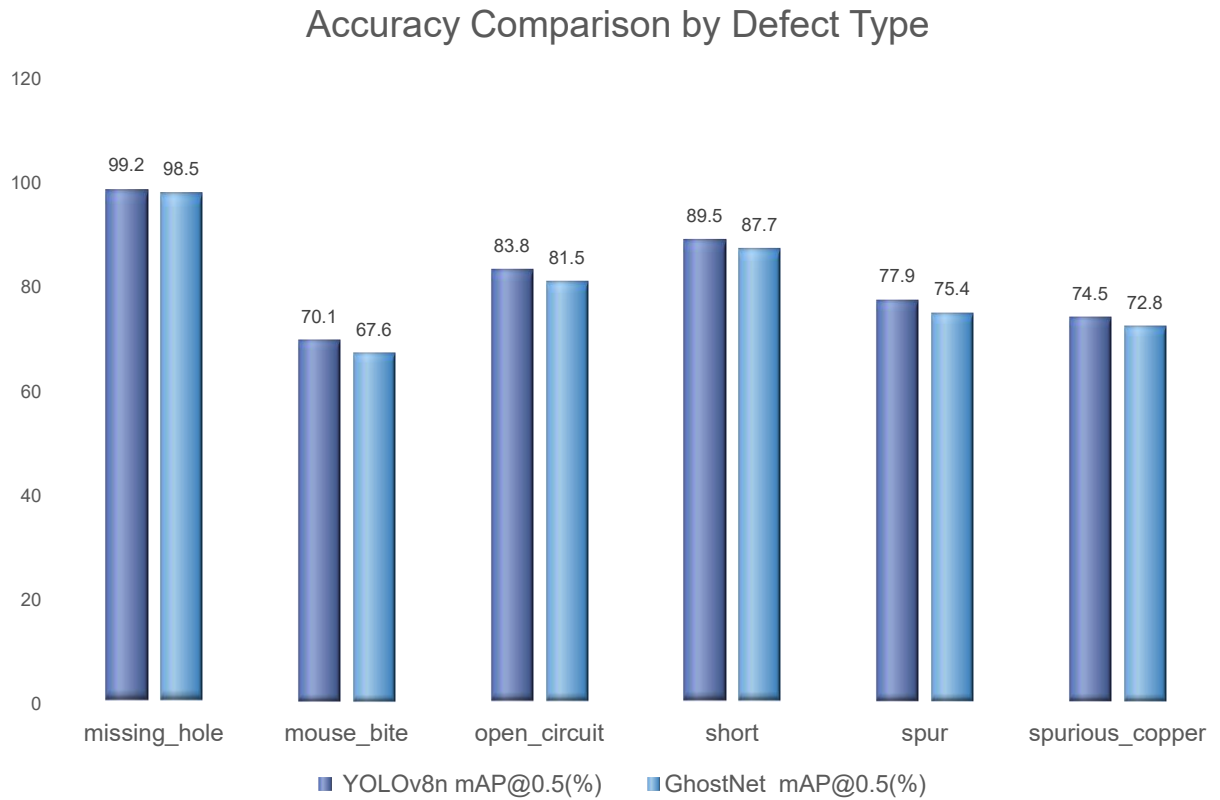


Figure 2.2.3 accuracy comparison by defect type

If we only look at Figure 2.2.3, the most obvious difference is obviously not the accuracy difference between YOLO and Ghostnet. The accuracy difference between different defects is much greater than that between different models.

This phenomenon indicates that in current PCB defect detection research, the physical characteristics of defects themselves are the dominant factor determining detection accuracy, while the influence of model structure differences is a secondary factor. The leakage is close to full marks on both models, not because the model has strong detection ability, but because the visual characteristics of the leakage itself are too distinct. Similarly, mouse bites scored only about 70 points in both models, because the random and blurry edges of mouse bites are difficult samples in the field of object detection. Lightweight transformation has not fundamentally changed this situation. Its impact on each category is generally uniform, although the fuzzy defect accuracy loss is slightly greater, the difference is very limited. Its impact on each category is generally uniform, and what was originally easy to detect remains easy, while what was originally difficult to detect remains difficult.

Although GhostNet's accuracy declines on some minor defects, it is almost negligible compared to the accuracy difference caused by the defects themselves. This also means that instead of investing too much energy in small optimizations of the model structure, it is better to focus more on defect types that are inherently more difficult to detect. This discovery points to a more cost effective optimization direction for future research.

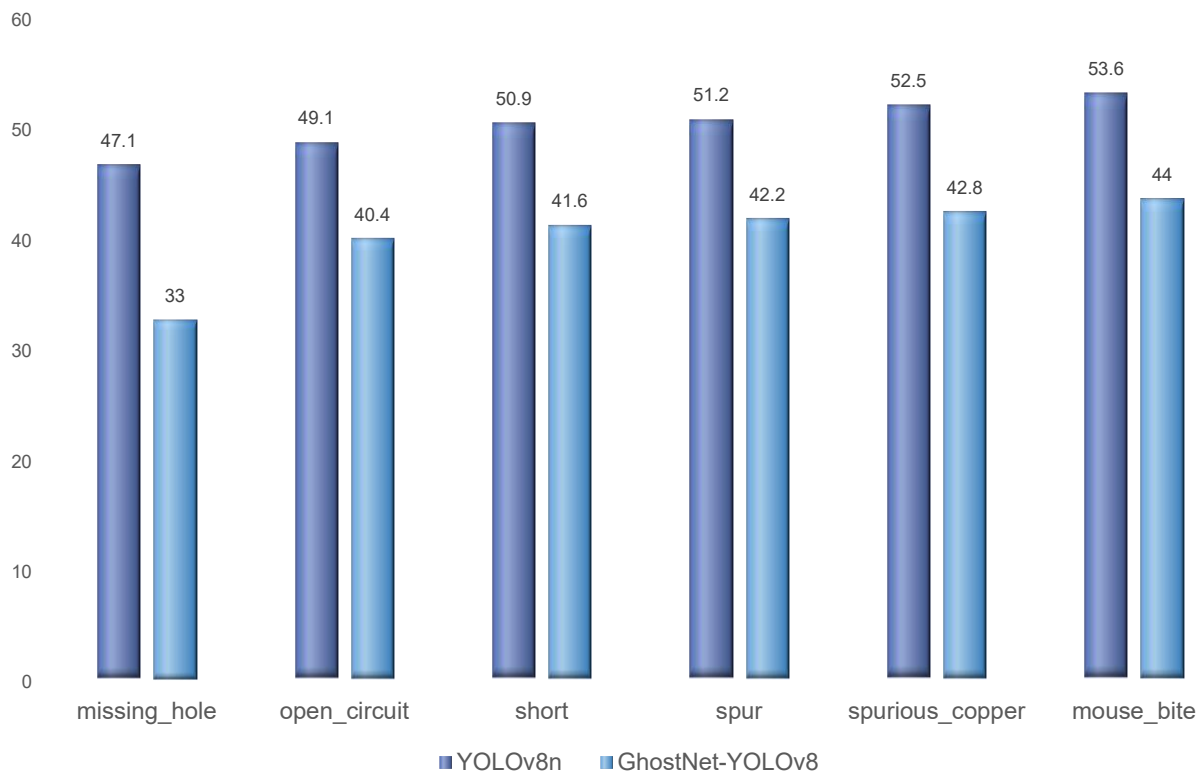


Figure 2.2.4 Comparison of speed improvement with different defects between YOLOV8 and Ghostnet

GhostNet-YOLOv8 achieves significant inference acceleration across all six defect categories. The inference time for the five types of defects, including open circuit, short circuit, burr, mixed copper, and mouse bite, is not significantly different. The single detection time of the original YOLOv8n model is between 49.1-53.6 ms, while GhostNet YOLOv8 reduces the time consumption to 40.4-44.0 ms. The absolute speed difference of the five types of defects is concentrated in the narrow range of 8.7-9.7 ms, with an average speed improvement of about 17.9%. The essential feature of Ghost convolution acceleration is to reduce the computational complexity of redundant standard convolutions in the backbone network. This reduction applies to the entire forward propagation process of each input image, regardless of the specific type of defects contained in the image.

However, the acceleration data for leak defects significantly deviates from this benchmark. Its relative acceleration ratio is as high as 30.0%, almost 1.7 times the average acceleration amplitude of the other five types of defects. This exceptionally prominent value requires careful interpretation, and we have considered two possible paths of explanation.

Leakage holes on the PCB surface are usually represented as a circular blank area with regular boundaries. The core design concept of Ghost convolution is to identify and remove redundant feature maps, and replace expensive standard convolutions with lower computational cost linear transformations. Therefore, when dealing with simple images such as leaks, the Ghost module may trigger deeper computational reduction. The network has already captured sufficient discriminative features in the shallow layers, and the computational workload in the deeper layers is further compressed, resulting in acceleration benefits beyond the average level. If this explanation holds true, then the high acceleration of leaky holes is not due to data bias, but rather Ghost convolution has a more significant acceleration on simple samples.

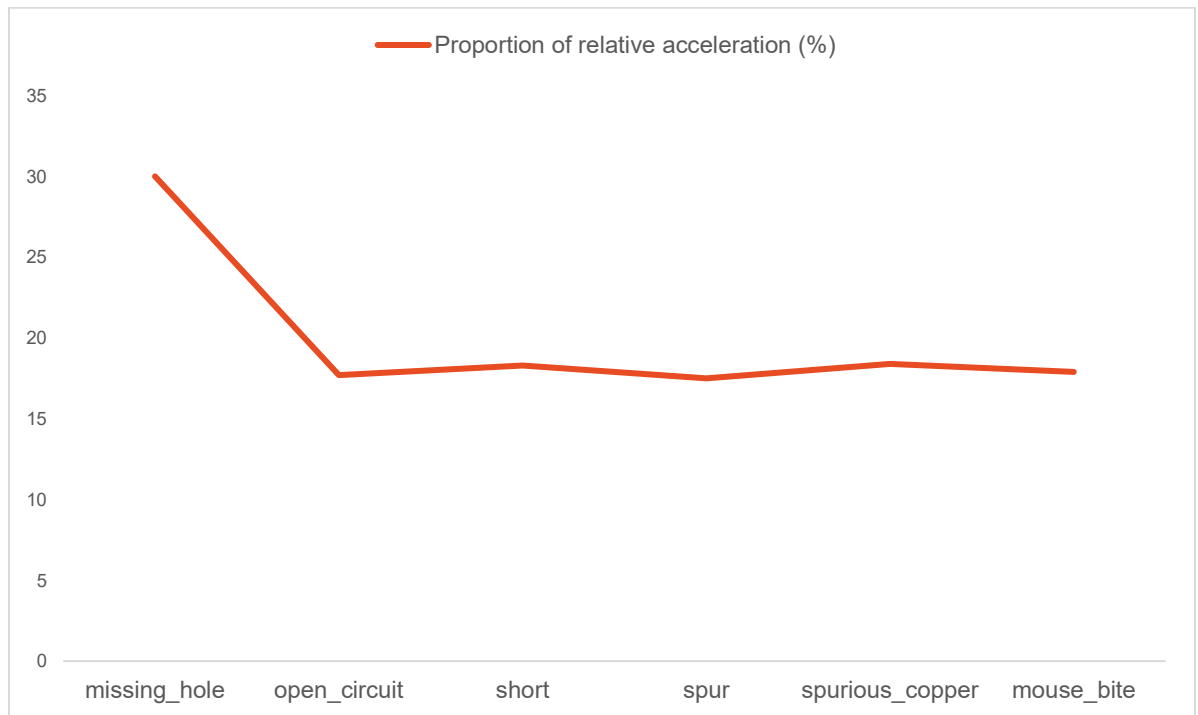


Figure 2.2.5 Proportion of relative acceleration with different defects between YOLOV8 and Ghostnet

The second possibility points to statistical bias at the dataset level. During the data annotation process, we noticed that leakage defects often cluster multiple defects on the same PCB board. When the total number of various defects in the dataset is similar, the number of images with missing holes in the dataset will be much smaller, about one-third of other defects. This resulted in the already limited number of leak defects being dispersed to a smaller number in the test set, with a total of 700 images. When divided into six types of defects, multiplied by 0.15 and then multiplied by one-third, there were only 6 images. This sample size is too small and has strong randomness and contingency. So the higher acceleration of leakage defects may not be due to their inherent properties, but rather to randomness.

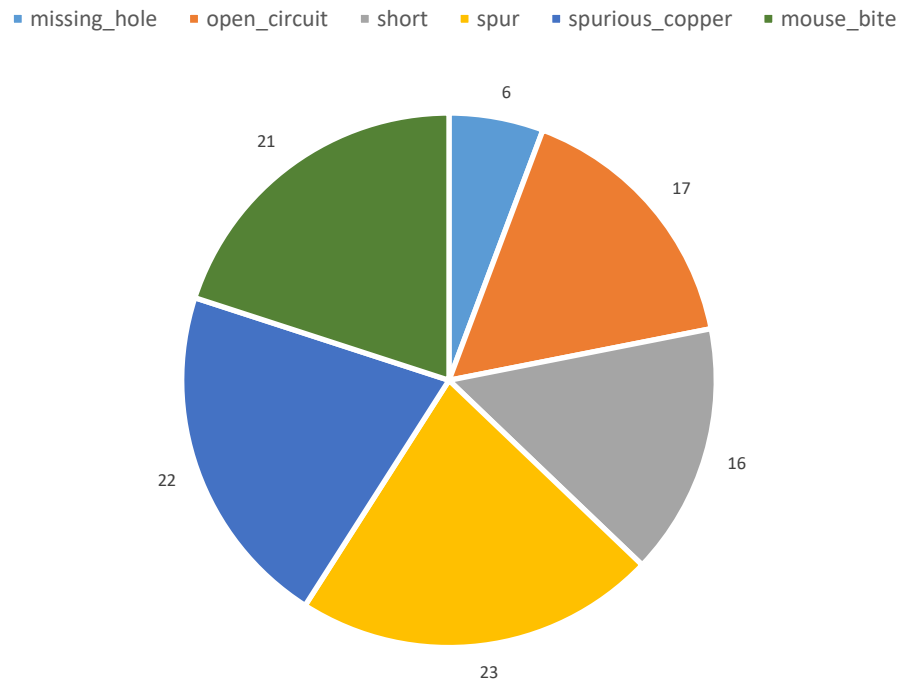


Figure 2.2.6 The number of images with different defects in the test set

Of course, it may also be affected by both situations simultaneously. To achieve more reliable results, a larger dataset is needed. This experiment is limited by hardware conditions, and a dataset of 700 images is basically the limit. Experiments on larger dataset can serve as a future research direction.

Chapter 3

Conclusion and Future Work

3.1 Conclusion

Starting from the practical needs of industrial edge deployment, this study designed and implemented a YOLOv8 lightweight PCB defect detection model based on GhostNet improvement, and analysed the experimental results from multiple perspectives. The core conclusions of this article can be summarised as follows.

The parameter count of Ghostnet's improved model has been reduced from 3.15M in the original YOLOv8n to 1.75M, with a decrease of 44.44%. The weight file size has been reduced from 6.2MB to 3.7MB, reducing storage space usage by 40.32%. The lightweight effect is significant. The average inference time for a single 640×320 PCB image has been reduced from about 41.9ms to about 33.6ms, with a speed increase of about 18%. The overall model mAP@0.5 only decreased from 82.3% to 80.6%, with an accuracy loss of less than 1.7 percentage points.

This kind of profit and loss is quite cost-effective in industrial applications. This also completes our initial idea: Ghost convolution achieves lightweighting by reducing redundant feature calculations, which can significantly reduce the model's dependence on hardware resources without significantly damaging the network's core expression ability. The accuracy loss of less than two percentage points in exchange for nearly half the compression of the model volume and nearly one-fifth the acceleration is practically attractive in industrial application scenarios.

In the subdivision testing of six typical defects, the differences between different defects are much greater than the differences between different models. The difference in detection

accuracy between different types of defects is nearly 30 percentage points. The mAP of the leakage defect is close to full marks in both models, while the mouse bite defect is only at a level of just over 70% in both models. Lightweight transformation will not fundamentally change the difficulty pattern determined by the physical characteristics of defects themselves. Defects that were originally easy to detect are still easy, and defects that were originally difficult are still difficult. Easy performances will not be too good, and difficult ones will not have too much loss.

The accuracy difference between GhostNet and YOLOv8n in different defect categories is at most 2.5 percentage points, while the accuracy gap between leaks and mouse bites is close to 30 percentage points. The core bottleneck currently faced in the field of PCB defect detection may not lie in the selection or fine-tuning of CNN architecture. Defects with small targets, blurry edges, and highly fused texture features and background are significantly more difficult to detect than other types. Using high-resolution input to preserve the detailed information of small targets or designing specialised feature enhancement strategies for small targets may result in more significant accuracy improvements at lower computational costs. This may be another possible direction for lightweighting improvement.

This certainly does not mean that the differences between model structures can be ignored. The impact of lightweight compression on different types of defects is not equal. On defects with relatively regular and clear contours, such as leaks and short circuits, the mAP difference between the two models is small, and the impact of lightweighting can be almost ignored. But in defects with irregular shapes and blurred edges, such as mouse bites and burrs, the accuracy difference exceeds 2.5%. This indicates that the physical characteristics of the defect itself determine the upper limit of detection difficulty, but the model structure affects the extent to which the detection can approach the upper limit. The lightweight compression of GhostNet has a significantly greater impact on small targets and weak feature defects than on highly recognisable defects. The missed detection situation indicated by the loss function has a more direct impact on the model's localisation ability,

especially its ability to judge fuzzy edges, while its classification and accurate discrimination abilities are basically preserved.

In terms of inference speed, the speed improvement ratio of five types of defects, including open circuit, short circuit, burr, copper impurity, and mouse bite, is highly consistent and stable within a narrow range of 17.5% -18.4%. This indicates that the computational reduction brought by GhostNet is systematic and stable across categories. The acceleration rate of leakage defects has reached 30%, significantly higher than other types. We discussed two possible explanations for this: one is that the visual features of leak images are extremely simple and vivid, and the Ghost module may trigger deeper computational reduction when processing such low complexity images. Another possibility is that the sample size for leak testing is relatively limited, and small sample deviations may, to some extent amplify the acceleration data. This article tends to consider around 18% as a more universally applicable stable acceleration level, as it is more in line with the system's automatic output of 20% single acceleration. The abnormal performance of the missing hole acceleration of 30% deserves further validation on a larger dataset.

From an engineering application perspective, I believe the positioning of GhostNet YOLOv8 is clear. Its small size and lower computational load allow it to run directly on embedded devices with limited computing power or on regular CPU environments. For small and medium-sized PCB production enterprises that hope to achieve automated quality inspection at lower hardware costs, the advantages of sufficient accuracy, small size, and fast speed in testing may be more practical than simply pursuing the highest accuracy of large models in laboratories or computing centres.

Compared with some existing lightweight detection schemes in the literature, this method does not introduce additional attention modules or complex multi-branch feature fusion structures. It purely relies on the efficiency optimisation of Ghost convolution at the feature generation level, and trades convenience for minimal accuracy cost. This simple technical route itself also reduces the complexity of debugging and maintenance of the model in

actual projects. Overall, this study confirms that the lightweight route based on GhostNet is feasible for PCB defect detection tasks and provides a reference technical idea for the marginalisation deployment of other similar industrial visual inspection tasks.

The main contribution of this study is the utilization of a practical lightweight framework. By replacing the YOLOv8n backbone network with GhostNet, this study validates a simple and effective lightweight improvement path for PCB defect detection, providing a deployable solution for resource constrained industrial edge devices. Unlike many existing studies that only focus on algorithm accuracy, this work explicitly links model design with two different processes: online screening and offline analysis. I believe that for production line deployment, the main goal is fast binary classification and localization, rather than detailed fine-grained types, which provides a more realistic and cost-effective direction for future research. This article also provides a detailed analysis of the types of defects caused by the impact of lightweight compression. The performance degradation caused by lightweighting is mainly manifested in the reduced recall rate for small and irregular defects, while the ability to locate and classify prominent defects is well preserved. In addition, we found that the inherent physical characteristics of defects themselves are more dominant in terms of detection difficulty compared to the choice between basic and lightweight models, which will guide future optimization work to improve difficult samples.

3.2 Future Work

During the process of this study, some unresolved problems with current methods were also exposed, which precisely pointed out possible directions for future work.

Firstly of all, defects with smaller areas and weaker features remain the biggest challenge in deep learning detection. After lightweighting, the detection accuracy of three defects, namely mouse bite, burrs, and copper impurities, decreased by 2.5, 2.5, and 1.7 percentage points, respectively. From the different performances of Box Loss, DFL Loss, and BoxR function in the two models, it can be seen that the root of the problem lies in the localisation stage rather than the classification stage. Lightweight compression of shallow fine-grained feature channels, which are more critical for locating small targets, results in the loss of some weak defect features in the early stages of feature extraction.

A possible improvement method is to add lightweight attention mechanisms with low computational consumption, such as coordinate attention or efficient channel attention, based on the Ghost module. These modules introduce a small number of additional parameters, and by guiding the model to focus limited attention resources on easily overlooked small defect areas, the accuracy can be improved by one or two percentage points without significantly increasing the computational burden. In addition, for highly irregular defects such as mouse bites, special feature enhancement structures can be designed. Targeted data augmentation strategies are also a direction worth exploring.

Secondly, the robustness of models in complex industrial environments still needs to be tested and improved. The dataset of this experiment has high image quality and a clear PCB surface. However, in real production workshops, there may be variations in lighting, the surface of the board may be contaminated with oil or impurities, and overexposure may occur locally due to reflections from the wiring. Lightweight models are more likely to misclassify environmental noise as defects when complex interference is superimposed due to the significant reduction of feature channels. Subsequent research can construct a

composite dataset containing multiple types of real noise. The stability and generalisation ability of the model in non-ideal situations need to be systematically tested and improved.

Further ultimate lightweighting and practical deployment are also pragmatic directions. At present, this study has compressed the model to the level of F1.75M parameters and 3.7MB storage volume. Conventional embedded platforms, such as NVIDIA Jetson Nano, are already friendly enough, but if it is to be further deployed on microcontrollers or low-cost FPGAs with more demanding computing power and storage, deeper compression is needed. INT8 weight quantisation may further compress the model volume and inference latency. If it can be completed without causing a significant drop in accuracy, the deployment threshold will be lowered by another order of magnitude.

In addition, the current model's functionality is still at the stage of detecting specific defects and their locations. In actual production lines, it is not enough to only know the location and category of defects. Operators and engineers also need to understand whether the size of the defect exceeds the standard, how severe it is, and which process may have experienced abnormalities. In the future, we can consider adding the ability to determine the specific situation of defects outside of the detection module.

In the past two years, visual transformers have demonstrated complementary capabilities to traditional CNNs in multiple visual tasks. The advantage of a lightweight CNN lies in its strong ability to extract local features and low computational complexity. The advantage of Transformer lies in its ability to model global dependencies and long-distance feature relationships. For defects with random shapes and discrete distributions in PCB images, the global perception of Transformer may precisely compensate for the natural limitations of CNN in local perception. In the future, we can try the fusion architecture of a lightweight CNN and a lightweight Transformer, and explore whether we can utilise the strengths of the Transformer to overcome the difficult-to-detect defects that still make pure CNN models challenging, while maintaining lightness. This may be an important evolutionary direction for the next generation of industrial visual lightweight inspection models.

In future work, we plan to design a two-stage collaborative detection framework. In the first stage, the GhostNet model will be deployed directly on the production line to perform fast and timely defect detection and rough localization. Then, the powerful model in the second stage will perform defect type and precise morphology analysis on the discovered defective boards to track process anomalies. A powerful offline model can serve as a teacher, further enhancing the ability of lightweight online models to identify difficult samples without increasing inference costs. This collaboration is consistent with real factory workflows and provides a possible direction for balancing real-time performance and deep analysis in industrial quality control systems.

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